

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

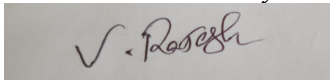
Annexure B
Analytical Problem Solving

<i>Criteria</i>	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I 192311037/Rajesh.v (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2- Assignment Tool -1

- (Q) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order)
- 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70
- What is the mean of the data? What is the median?
 - What is the mode of the data? comment on data's modality
 - What is the midrange of the data
 - Can you find the first quartile ^(Q1) & the third quartile of data (Q3)
 - Give the five number summary of the data
 - Show a boxplot of the data

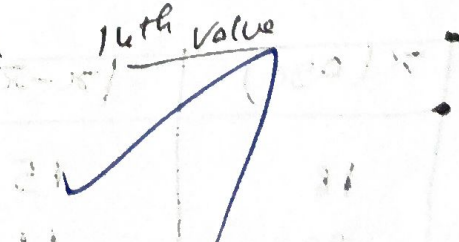
Sol

a) Mean = $\frac{1}{n} \sum_{i=1}^n x_i$

$$= \frac{809}{24} = 29.96 \approx 30$$

Median = $\frac{n+1}{2} = \frac{24+1}{2} = 14$ 14th value

= 25



- b) 25 and 35 are coming repeatedly
As only two value are repeating which is bimodal

c) Midrange = $\frac{\text{max value} + \text{min value}}{2} = \frac{70+13}{2} = 41.5$

d) Q1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

Median = $\frac{n+1}{2} = \frac{13+1}{2} = 7^{\text{th}}$ value

$$Q3 = 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 50, 70$$

$$\text{Median} = \frac{13+1}{2} = \frac{14}{2} = 7^{\text{th}} \text{ value}$$

$$= 35$$

e) Five num summary = Min, Q1, Median, Q3, Max

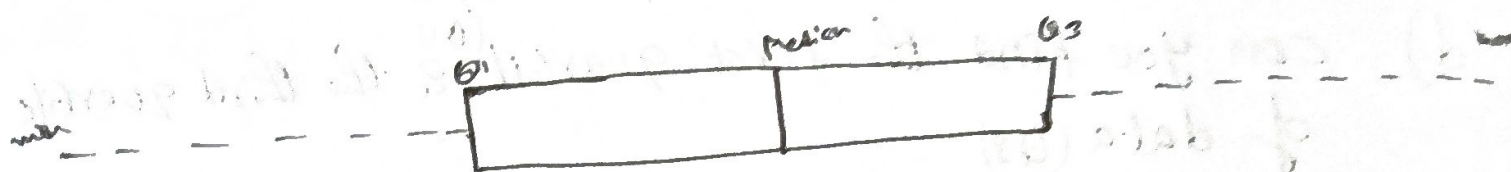
$$\text{Min} = 13$$

$$Q1 = 20$$

$$\text{Max} = 70$$

$$Q3 = 35$$

f) Box Plot



13 15 16 19 20 21 22 25 30 33 35 36 40 45 46 50 70

2) Given the Following measurements for the variable age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28. Standardize the variable by the following: compute the mean absolute deviation of age

Sol

$$\text{Mean} = \frac{1}{n} \sum_{i=1}^n x_i \Rightarrow \frac{330}{10} = 33$$

$x(\text{age})$	$ x - \bar{x} $
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5
88	

$$\text{Mean Absolute Deviation} = \frac{\sum |x - \bar{x}|}{n}$$

$$= \frac{88}{10}$$

$$= 8.8$$

3) SUPP

Q3

13, 15

35,

Use

Ans

sl Me

Bin

col

Bin

B1 13, 15

B2 16, 19

B3 20, 22

B4 22, 25

B5 25, 28

B6 33, 35

B7 35, 40

B8 36, 45

B9 46, 50

B10 46, 70

Smoothed val

a) Use

group

a) m

m

b) z

Q3) Suppose that the data for analysis includes the attributes age. The age values for the data tuples are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Use smoothing by bin means to smooth the above data using a bin depth of 3. Illustrate your steps

Sol

$$\text{Mean} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{809}{27} = 29.96 \approx 30$$

Bin depth = 3

count of bin = $\frac{\text{occurrence}}{\text{bin depth}} \Rightarrow \frac{27}{3} = 9$

Bin	Mean	Smoothed values
B1 13, 15, 16	14.67 14.67	14.67, 14.67, 14.67
B2 16, 19, 20	18.33	18.33, 18.33, 18.33
B3 20, 21, 22	21	21, 21, 21
B4 22, 25, 25	24	24, 24, 24
B5 25, 25, 30	26.67	26.67, 26.67, 26.67
B6 33, 33, 35	33.67	33.67, 33.67, 33.67
B7 35, 35, 35	35	35, 35, 35
B8 36, 40, 45	40.33	40.33, 40.33, 40.33
B9 46, 52, 70	56	56, 56, 56

Smoothed values are replaced the mean for all the 3 values

Q4) Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

a) min-max normalization by setting min = 0 and max = 1

b) Z-score normalization

Sol a) Min-max normalization

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x' = \frac{x - 200}{1000 - 200} \Rightarrow \frac{x - 200}{800}$$

Original Value	Calculation	Normalized Value (x')
200	$(200 - 200) / 800$	0.000
300	$(300 - 200) / 800$	0.125
400	$(400 - 200) / 800$	0.250
600	$(600 - 200) / 800$	0.500
1000	$(1000 - 200) / 800$	1

$$x = [0, 0.125, 0.250, 0.500, 1]$$

b) Z-Score normalization

$$z' = \frac{x - \mu}{\sigma}$$

$$\text{Mean } (\mu) = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$$

Standard deviation

x	$x - \mu$	$(x - \mu)^2$
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	250,000
		4,00,000

$$\sigma^2 = \frac{400000}{5} = 80,000$$

$$\sigma = \sqrt{80000} \\ = 282.84$$

Z Score

Original

$$Z = \frac{x - \mu}{\sigma}$$

Z Score

200

$$-300/282.84$$

$$-1.06$$

300

$$-200/282.84$$

$$-0.71$$

400

$$-100/282.84$$

$$-0.35$$

600

$$100/282.84$$

$$0.35$$

1000

$$500/282.84$$

$$1.767$$

$$Z \text{ Scores} = [-1.06, -0.71, -0.35, 0.35, 1.77]$$

⑤ Suppose a group of 12 Sales Price records has been Sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three bins by each of the following methods

- equal-frequency Partitioning
- equal-width Partitioning
- clustering

Sol No. of records = 12

no. of bins = 3

a) Records per bin = $\frac{12}{3} = 4$

Bin

Values

B1

5, 10, 11, 13

B2

15, 35, 50, 55

B3

55, 72, 92, 204, 215

~~B4~~

b) Equal width Partitioning

$$\text{width} = \frac{\text{max} - \text{min}}{\text{no. of bins}} = \frac{215 - 5}{3} = \frac{210}{3} = 70$$

Intervals

Bin1 : 5-74

Bin2 : 75-144

Bin3 : 145-215

Bin

Interval

Values

Bin1

5-74

5, 10, 11, 13, 15, 35, 50, 55, 72

Bin2

75-144

92

Bin3

145-215

204, 215

⇒ Clustering Partition

Cluster(Bin)

Values

cluster1

5, 10, 11, 13, 15

cluster2

35, 50, 55, 72, 92

cluster3

204, 215

Q Suppose a hospital tested the age and body fat data for 18 randomly selected adults with following

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

a) calculate the...

b) Draw the box plot

c) Draw a scatter

d) Normalize the

e) calculate the variable Positive

SD

Age

a) Mean($\bar{x}$) = $\frac{1}{n} \sum_{i=1}^n x_i$

Median = $\frac{n+1}{2}$

fat%

Mean($\bar{y}$) = $\frac{1}{n} \sum_{i=1}^n y_i$

Median = $\frac{n+1}{2}$

Mode = 23, 27, 54, 58

$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$

Mode for fat = no mode

$S = 9.61\%$

b) Age

Min = 23, Max =

Q1 (lower half) = 23

Median = 39

Q3 (upper half) :

Median = 5

23 27 39 41 4

min - - - - -

- calculate the mean, median, & standard deviation of age & %fat
- Draw the boxplots for age & %fat (on a separate page)
- Draw a scatter plot and a 2-2 Plot based on these variables
- Normalize the two variables based on z-score normalization
- calculate the correlation coefficient - Are these two variables positively or negatively correlated?

5d) Age

$$a) \text{Mean}(\bar{x}) = \frac{1}{n} \sum_{i=1}^n x_i = \frac{836}{18} = 46.4$$

$$\text{Median} = \frac{n+1}{2} = \frac{19}{2} = 9.5 \quad \frac{50+52}{2} = 51$$

%fat

$$\text{Mean}(\bar{y}) = \frac{1}{n} \sum_{i=1}^n y_i = \frac{518.1}{2} = 259.05$$

$$\text{Median} = \frac{n+1}{2} = \frac{30.2+31.2}{2} = 30.7$$

$$\text{Mode} = 23, 27, 54, 58$$

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = 13.74$$

$$\text{Mode for fat} = \text{no mode}$$

$$S = 9.61\%$$

b) Age

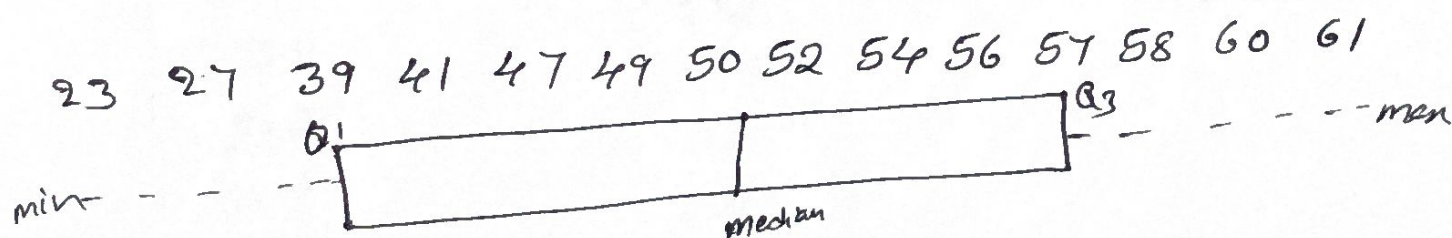
$$\text{Min} = 23, \text{Max} = 61$$

$$Q_1 (\text{lower half}) = 23, 23, 27, 27, 39, 41, 47, 49, 50$$

$$\text{Median} = 39$$

$$Q_3 (\text{upper half}) : 52, 54, 54, 56, 57, 58, 58, 60, 61$$

$$\text{Median} = 57$$



ABOX)

$\frac{x-\mu}{\sigma}$

$\left(\frac{x-\mu}{\sigma}\right)^2$

23	23-46.44/13.74	-1.71
23	23-46.44/13.74	-1.71
27	27-46.44/13.74	-1.41
27	27-46.44/13.74	-1.41
39	39-46.44/13.74	-0.54
41	41-46.44/13.74	-0.40
47	47-46.44/13.74	0.04
49	49-46.44/13.74	0.19
50	50-46.44/13.74	0.26
52	52-46.44/13.74	0.41
54	54-46.44/13.74	0.55
54	54-46.44/13.74	0.55
56	56-46.44/13.74	0.70
57	57-46.44/13.74	0.77
58	58-46.44/13.74	0.84
58	58-46.44/13.74	0.84
60	60-46.44/13.74	0.99
61	61-46.44/13.74	1.06

Z-SCORE

Bodyfat (%)

Z-SCORE

9.5	-2.01
26.5	-0.24
7.8	-2.18
17.8	-1.14
31.4	0.27
25.9	-0.30
27.4	-0.14
27.2	-0.16
31.2	0.25
34.6	0.61
42.5	1.43
28.8	0.00
33.4	0.48
30.2	0.15
34.1	0.55
32.9	0.43
41.2	1.29
35.7	0.72

7) Fat

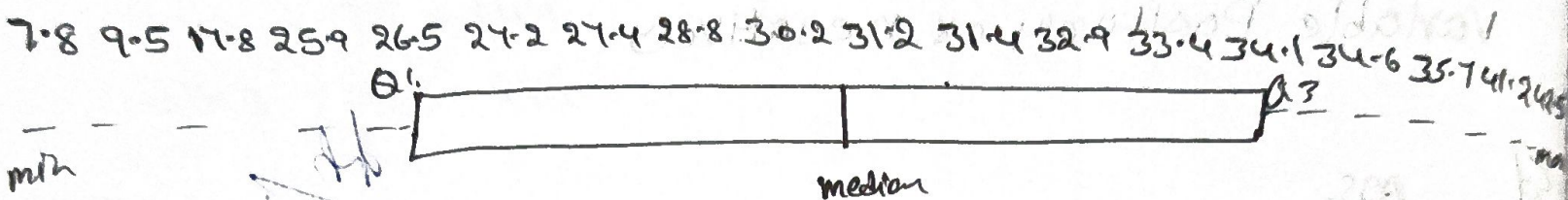
min = 7.8, max = 42.5

Q1 (Lower half) = 7.8, 9.5, 17.8, 25.9, 26.5, 27.2, 27.4, 28.8, 30.2

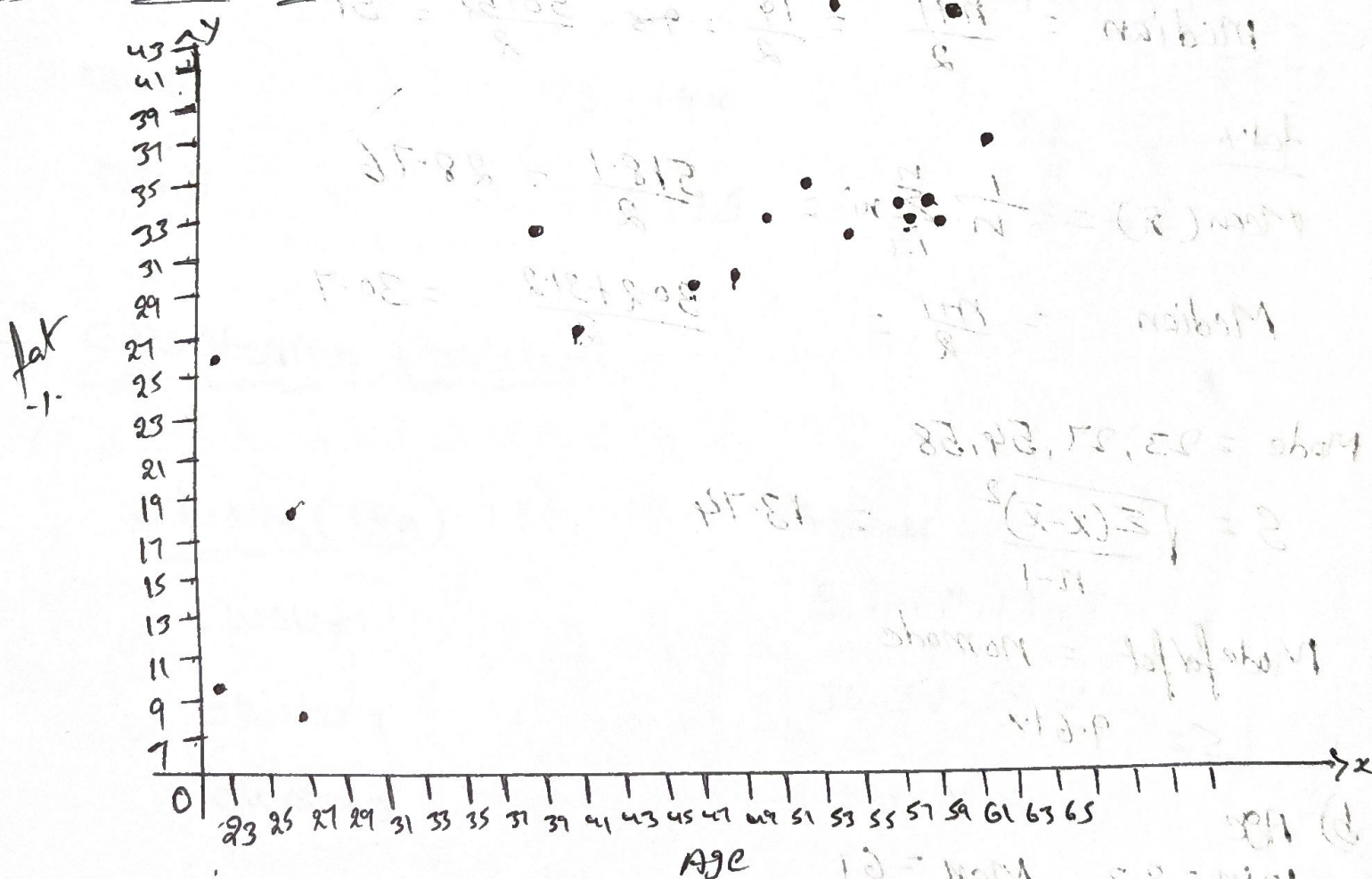
median = 26.5

Q3 (Upper half) = 31.2, 31.4, 32.9, 33.4, 34.1, 34.6, 35.7, 41.2, 42.5

median = 34.1



8) Scatter Plot



9) Z-score

$$Z = \frac{x - \mu}{\sigma}$$

$$\mu = 28.78$$

$$\sigma = 9.61$$

$$\text{Mean} = 28.78$$

$$\sigma = 9.61$$



ABW

23

23

27

27

39

41

47

49

50

52

54

54

56

57

58

58

60

61

Z-score

Bodyfat

9

26

7

17

31

25

27

31

34

42

28

32

30

34

32

41

35

x	y	x	y
23	9.5	529	90.25
23	26.5	529	702.25
27	7.8	729	60.84
27	17.8	729	316.84
39	31.4	1521	985.96
41	25.9	1681	670.81
47	27.4	2209	750.76
49	27.2	2401	739.84
50	31.2	2500	973.44
52	34.6	2704	1197.16
54	42.5	2916	1806.25
54	28.8	2916	829.44
56	33.4	3136	1115.56
57	30.2	3249	912.04
58	34.1	3364	1162.81
58	32.9	3364	1082.41
60	41.2	3600	1697.44
61	35.7	3721	1274.49
836	518.1	41144	16368.59
		41798	25323.2

$$r = \frac{18(25323.2) - (836)(518.1)}{\sqrt{[18(41798 - 698,896)] [18(16368.59) - 268,427.61]}}$$

$$= \frac{455817.6 - 433131.6}{\sqrt{18(-657098) [18(-252,059.00)]}}$$

$$= \approx 0.82$$

As the correlation coefficient value ≈ 0.82 that is near 1 which represent both coefficients are positively correlated.